THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 1

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In a right triangle with perpendicular 4, base 3 and hypotenuse 5, $\sin\theta =$ 1

(a) $\frac{4}{3}$ **(b)** $\frac{4}{5}$ **(c)** $\frac{5}{4}$ **(d)** $\frac{3}{5}$

2. From a point on level ground, 7 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. The height of the tower is: 1

(a) 14 m **(b)** 7 m **(c)** $\frac{7\sqrt{3}}{3}$ m **(d)** $7\sqrt{3}$ m

3. A point P is 5 cm from the centre O of a circle of radius 3 cm. The length of the tangent drawn from P to the circle is: 1

(a) 4 cm **(b)** 2 cm **(c)** 16 cm **(d)** 8 cm

4. The radius of a circle is 7 cm. The area of the circle is: 1

(a) 44 cm² **(b)** 154 cm² **(c)** 22 cm² **(d)** 308 cm²

5. In a right triangle with perpendicular 4, base 3 and hypotenuse 5, $\tan\theta =$ 1

(a) $\frac{3}{5}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{4}{3}$ **(d)** $\frac{4}{5}$

6. From a point on level ground, 5 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $10\sqrt{3}$ m **(b)** $\frac{5\sqrt{3}}{3}$ m **(c)** $5\sqrt{3}$ m **(d)** 5 m

7. The length of the tangent from an external point P to a circle is 8 cm and P is 10 cm from the centre. The radius of the circle is: 1

(a) 18 cm **(b)** 2 cm **(c)** 6 cm **(d)** 9 cm

8. The radius of a circle is 7 cm. The circumference of the circle is: 1

(a) 154 cm **(b)** 88 cm **(c)** 44 cm **(d)** 22 cm

9. The value of $\sin 30^\circ$ is: 1

(a) $\frac{1}{2}$ **(b)** 2 **(c)** 1 **(d)** 0

10. From a point on level ground, 9 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) $3\sqrt{3}$ m **(b)** $9\sqrt{3}$ m **(c)** 9 m **(d)** 3 m

11. A tangent to a circle of radius 7 cm has length 24 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 17 cm **(b)** 50 cm **(c)** 31 cm **(d)** 25 cm

12. The circumference of a circle is 44 cm. The radius of the circle is: 1

(a) 22 cm **(b)** 7 cm **(c)** 14 cm **(d)** 3.5 cm

13. The value of $\sin^2 20^\circ + \cos^2 20^\circ$ is: 1

(a) 0 **(b)** 20 **(c)** 1 **(d)** 2

14. A vertical pole is 9 m high. The angle of elevation of the sun is 45° and the pole casts a shadow on level ground. The length of the shadow is: 1

(a) $9\sqrt{3}$ m **(b)** $\frac{9}{2}$ m **(c)** 18 m **(d)** 9 m

15. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 3 cm. The length of each tangent is 4 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 7 cm **(b)** 16 cm **(c)** 8 cm **(d)** 14 cm

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. In a right triangle the perpendicular is 4 and the hypotenuse is 5. Find $\sin\theta$. 2

17. From a point on level ground, 10 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. Find the height of the tower. 2

18. A point P is 25 cm from the centre O of a circle of radius 20 cm. Find the length of the tangent drawn from P to the circle. 2

19. The radius of a circle is 3.5 m. Find the area of the circle. 2

20. Find the value of $sin30^\circ + cos60^\circ$. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. In $\triangle ABC$, right-angled at B, $AB = 3$ cm and $BC = 4$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

22. A ladder 12 m long leans against a vertical wall and makes an angle of 60° with the level ground. Find the height reached on the wall and the distance of the foot of the ladder from the wall (in this order). 3

23. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 12 cm. The length of each tangent is 16 cm. Find the perimeter of the quadrilateral $PAOB$. 3

24. A sector of a circle of radius 21 cm has central angle 45°. Find the length of its arc. 3

OR

If $\sin A = \frac{4}{5}$, find $\cos A$ and $\tan A$. 3

25. From the top of a cliff $10\sqrt{3}$ m high, the angle of depression of a boat on the sea is 60°. Find the horizontal distance of the boat from the foot of the cliff. 3

OR

Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 3 cm. The length of each tangent is 4 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. If $\tan\theta = \frac{4}{3}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 5

OR

Two points on level ground lie on opposite sides of a tower of height $6\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. Find the distance between the two points. 5

27. Tangents PA and PB are drawn from a point P at a distance of 5 cm from the centre O of a circle of radius 3 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

A circular park of radius 14 m is surrounded by a path 7 m wide. Find the area of the path. 5

END OF QUESTION PAPER **Total: 50 Marks**

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